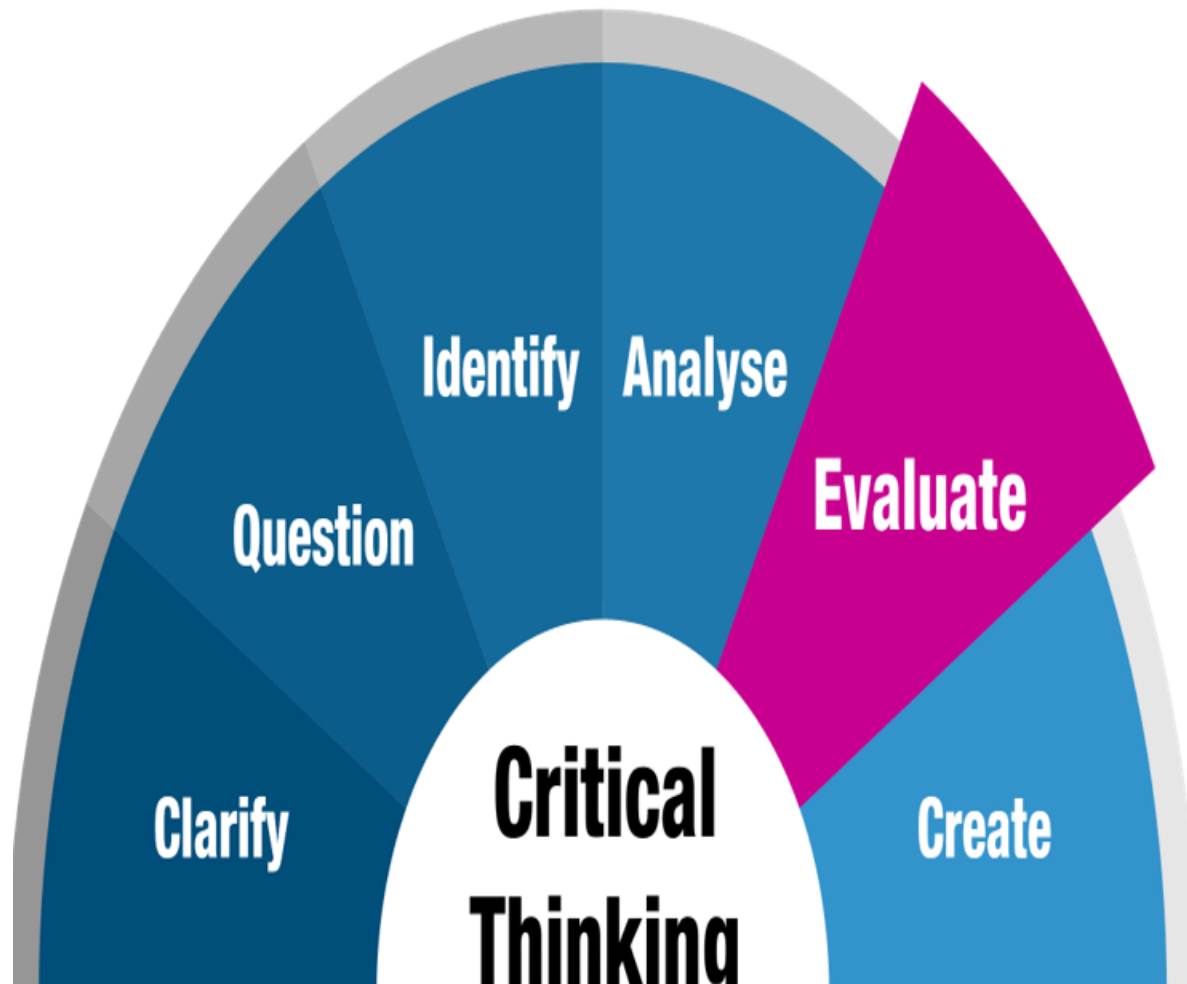




Evaluating Arguments

Evaluating Inductive Arguments: Strength & Cogency

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Evaluating Inductive Arguments: Strength, Truth, and Cogency

- Induction and Strength (strong , weak)

- *strong inductive argument:*

it is an argument ,if the premises are assumed true, it is improbable for the conclusion to be false.

In such arguments, the conclusion follows probably from the premises.

Evaluating Inductive Arguments: Strength, Truth, and Cogency

- *weak inductive argument*

It is an argument ,if the premises are assumed true, it is probable for the conclusions to be false.

In these arguments, the conclusion does not follow probably from the premises, even though it is claimed to.

Let's practice



Example-1:	Example-2:
(p1).This barrel contains one hundred apples. (p2).Eighty apples selected at random were found tasty. (c).Therefore, probably all one hundred apples are tasty.	(p1). This barrel contains one hundred apples. (p2). Three apples selected at random were found tasty. (c). Therefore, probably all one hundred apples are tasty.
Type of argument: <u>Strong inductive</u>	Type of argument : <u>Weak inductive</u>
Reason :because the conclusion actually follows probably from the premises.	Reason :because the conclusion does not actually follow probably from the premises, even though it is claimed to.

STRONG

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WEAK



please note

- The procedure for testing the strength of inductive arguments runs parallel to the procedure for deduction.
- **Inductive Argument Indicator Words**
- probably
- likely
- there's a good chance that
- in all likelihood
- tends to



please note

- Inductive reasoning is probabilistic in nature, meaning that the conclusion is not guaranteed to be true even if the premises are true.
- The strength of an inductive argument depends on the relevance and number of instances that support the conclusion.

Trick or Treats



Choose the correct answer

1. All inductive arguments involve generalizations to the conclusion and thus involve risk and probability.

- True
- False

2. All deductive arguments provide conclusive evidence for their conclusions.

- True
- False

• Answer

- 1. true 2. false



Choose the correct answer

1. All inductive arguments provide the best available evidence for their conclusions.

- True
- False

2. All inductive arguments involve risk and have uncertain conclusions, but some are stronger than others.

- True
- False

- Answer
- 1. false 2. true



Induction and Cogency

- **Cogent argument**

It is an inductive argument that is strong and has all true premises.

Cogent Argument = A strong argument + All true premises

- **Un-cogent argument**

It is an inductive argument that is either strong with one or more false premises, or weak, or both.

Argument terminology

